

AI UX Audit Agent - AuditPilot

Overview

AI UX Audit Agent is an agentic web auditing platform that evaluates a website across accessibility, performance, SEO, content quality, and competitive positioning. Users enter a website URL and receive a professional audit report with prioritized findings and actionable recommendations.

The goal is to automate the type of website review typically performed by UX consultants, accessibility specialists, SEO analysts, and product designers, while presenting results in a single dashboard and downloadable report.

Problem Statement

Website audits are often fragmented across multiple tools and require expertise in accessibility, performance optimization, UX writing, and SEO. Small businesses, startups, and agencies spend significant time gathering insights from different platforms before producing recommendations.

AI UX Audit Agent consolidates these activities into a single workflow powered by specialized AI agents.

User Workflow

1. User enters a website URL.
 2. The system crawls and analyzes the website.
 3. Multiple specialized agents run in parallel.
 4. Findings are aggregated and prioritized.
 5. Results are displayed in an audit dashboard.
 6. User exports a professional PDF report.
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System Architecture

Frontend (React)



FastAPI Backend



Audit Orchestrator

├— Accessibility Agent

├— Performance Agent

├— Copy Agent

├— SEO Agent

└— Competitor Agent



Playwright • Lighthouse • Gemini

Agent Responsibilities

1. Accessibility Agent

Purpose

Evaluate compliance with accessibility best practices and identify potential WCAG violations.

Tools

- Playwright
- Custom accessibility checks

Evaluations

- Missing image alt text
- Form labels
- Button accessibility
- Heading hierarchy
- ARIA attributes

- Keyboard accessibility
- Color contrast issues

Example Output

- High Severity: 24 images missing alt text.
 - Medium Severity: Multiple form fields missing labels.
 - Recommendation: Add descriptive alternative text and associated labels.
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2. Performance Agent

Purpose

Measure website performance and identify optimization opportunities.

Tools

- Lighthouse

Evaluations

- Performance score
- Largest Contentful Paint (LCP)
- Interaction to Next Paint (INP)
- Cumulative Layout Shift (CLS)
- Mobile responsiveness
- Image optimization

Example Output

- Performance Score: 62/100
 - Critical Issue: LCP exceeds recommended threshold.
 - Recommendation: Optimize hero images and reduce render-blocking resources.
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3. Copy Agent

Purpose

Evaluate website messaging clarity and communication effectiveness.

Tools

- Gemini

Evaluations

- Value proposition clarity
- Readability
- Marketing jargon
- Trust signals
- Call-to-action quality
- Content structure

Example Output

Finding:

The homepage headline does not clearly communicate the product's purpose.

Recommendation:

Replace generic marketing language with a specific value proposition that explains what the product does and who it serves.

4. SEO Agent

Purpose

Review on-page SEO fundamentals.

Tools

- Playwright
- Metadata extraction
- Gemini

Evaluations

- Title tags
- Meta descriptions
- H1 structure
- Heading hierarchy
- Open Graph metadata
- Structured data
- Image accessibility and indexing readiness

Example Output

- Missing meta description.
- Multiple H1 elements detected.

- Recommendation: Add descriptive metadata and improve document structure.
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5. Competitor Agent

Purpose

Provide market context by comparing the website with competitors.

Tools

- Gemini
- Search APIs
- Web scraping

Evaluations

- Competitor messaging
- Headline comparisons
- CTA comparisons
- Feature positioning
- Differentiation opportunities

Example Output

Finding:

Competitors communicate product value more explicitly within the first screen.

Recommendation:

Revise hero copy to clearly explain the product's primary benefit.

Dashboard Features

Overview Metrics

- Overall Audit Score
- Accessibility Score
- Performance Score
- Copy Quality Score
- SEO Score

Detailed Findings

Each category contains:

- Severity level
- Issue description
- Evidence
- Recommended action

Visual Components

- Score cards
 - Issue breakdowns
 - Priority rankings
 - Progress indicators
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PDF Report

The system generates a downloadable report containing:

Executive Summary

Overall assessment and key findings.

Accessibility Review

WCAG-related issues and recommendations.

Performance Review

Core Web Vitals and optimization opportunities.

Copy Review

Messaging, readability, and conversion recommendations.

SEO Review

Technical and content-related SEO observations.

Competitive Analysis

Positioning comparison and strategic recommendations.

Action Plan

Prioritized improvements based on impact and effort.

Technology Stack

Frontend

- React
- TypeScript
- Tailwind CSS
- shadcn/ui

Backend

- FastAPI
- Python

AI

- Gemini 2.5 Flash

Automation

- Playwright

Analysis

- Lighthouse

Storage

- SQLite (MVP)
- PostgreSQL (future)

Reporting

- ReportLab or PDFKit
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Future Enhancements

AI Redesign Suggestions

Generate improved page copy and layout recommendations.

Screenshot-Based Analysis

Analyze screenshots and identify visual design issues.

Conversion Optimization Agent

Evaluate CTAs, forms, and user flows.

UX Heuristic Evaluation Agent

Perform heuristic reviews using Nielsen usability principles.

Historical Monitoring

Track website improvements over time and generate comparison reports.

Portfolio Value

This project demonstrates:

- Agentic AI workflows
- Multi-agent orchestration
- LLM integration
- Web automation
- Accessibility evaluation
- UX research principles
- SEO analysis
- Full-stack software engineering
- Product design thinking

The project combines AI engineering with human-centered design and reflects real-world consulting workflows used by UX agencies, accessibility specialists, and digital product teams.